MNIST CNN Image Classifier Report

1. Objective

The objective of this project is to design and implement a Convolutional Neural Network (CNN) to classify handwritten digits (0–9) using the MNIST dataset. The model is also tested using custom handwritten images to evaluate real-world performance.

2. Dataset Description

The MNIST dataset consists of 60,000 training images and 10,000 testing images. Each image is a 28x28 grayscale image representing digits from 0 to 9.

3. Data Preprocessing

Data preprocessing is an important step to improve model performance:
- Pixel values were normalized from range (0–255) to (0–1).
- Images were reshaped to (28, 28, 1) to fit CNN input requirements.
- Labels were converted into one-hot encoded format.

This ensures consistency and helps the model learn efficiently.

4. Model Architecture

The CNN model includes:
- Convolutional layers to extract features.
- MaxPooling layers to reduce dimensionality.
- Dropout layers to prevent overfitting.
- Dense layers for classification.
- Softmax output layer for multi-class classification.

This architecture helps in capturing spatial features effectively.

5. Model Training

The model was trained for 5 epochs using the Adam optimizer and categorical crossentropy loss function. A validation split was used to monitor model performance during training.

6. Results

The model achieved a test accuracy of 99.06%, which exceeds the required 98%. This indicates strong performance and good generalization.

7. Custom Image Testing

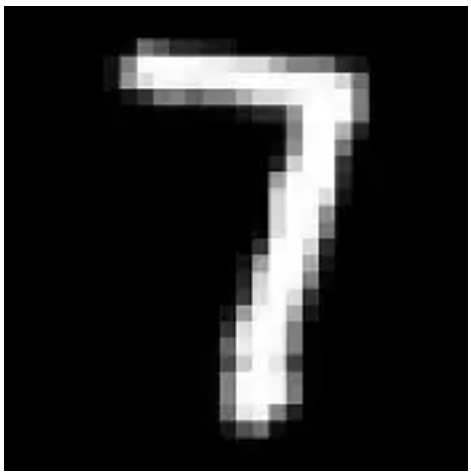
Two types of custom images were tested:
1. Black background with white digit: The model correctly predicted the digit.
2. White background with black digit: The model initially misclassified the digit.

After applying color inversion (255 - image), the prediction became correct. This highlights the importance of matching input format with training data.

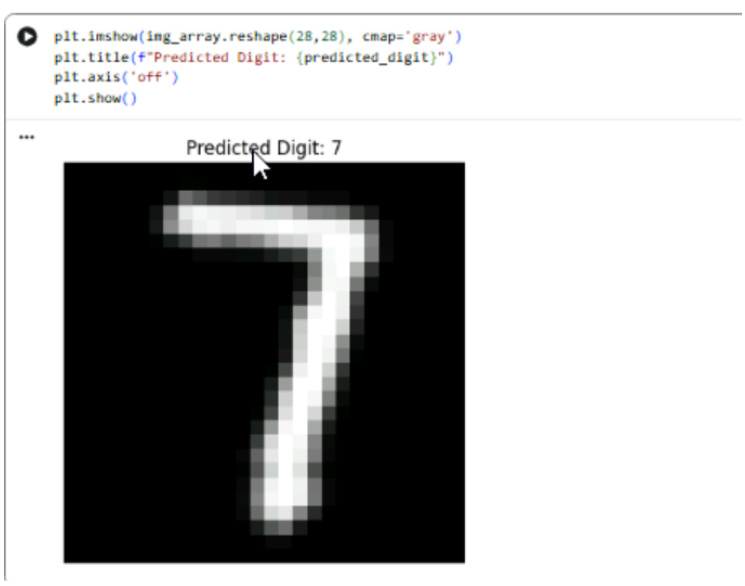
8. Input and Output Images

Insert input and predicted output images here for demonstration.

Input image-1



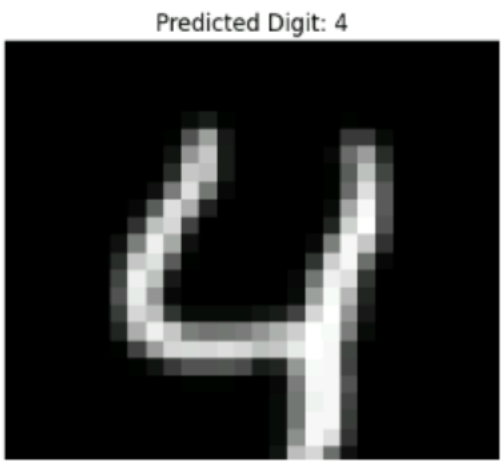
Output image-1



Input image-2

Output image-2

```
plt.imshow(img_array.reshape(28,28), cmap='gray')
plt.title(f"Predicted Digit: {predicted_digit}")
plt.axis('off')
plt.show()
```



9. Conclusion

The CNN model successfully classifies handwritten digits with high accuracy. Preprocessing plays a crucial role, especially when working with custom images. The project demonstrates the effectiveness of CNNs in image classification tasks.